

Uber Final Project
SPSS Models

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SPSS Model 1 (Slide 10): Paired Samples t-Test

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	What is your overall perception of brand (Uber)? - Brand Perception	3.54	76	1.051	.121
	On a scale from 0-10, how would you rate your most recent Uber ride?	7.26	76	2.340	.268

Paired Samples Correlations

		N	Correlation	Significance	
				One-Sided p	Two-Sided p
Pair 1	What is your overall perception of brand (Uber)? - Brand Perception & On a scale from 0-10, how would you rate your most recent Uber ride?	76	.717	<.001	<.001

Paired Samples Test

		Paired Differences				Significance			
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference		t	df	
					Lower	Upper			
Pair 1	What is your overall perception of brand (Uber)? - Brand Perception - On a scale from 0-10, how would you rate your most recent Uber ride?	-3.724	1.748	.201	-4.123	-3.324	-18.570	75	<.001

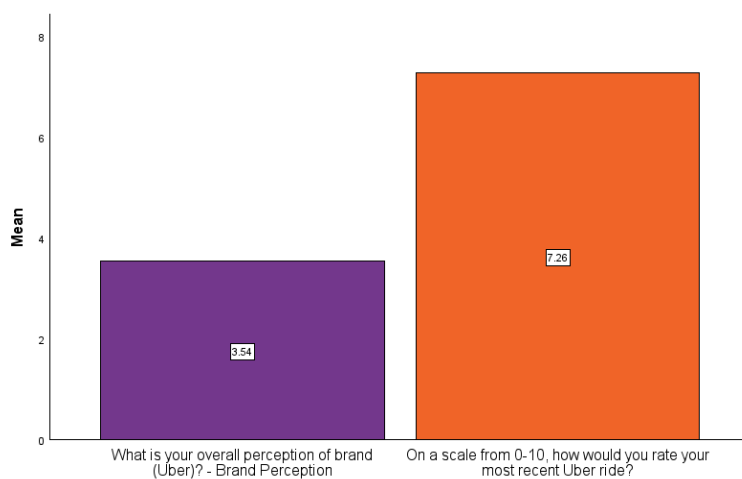
Paired Samples Effect Sizes

		Standardizer ^a		95% Confidence Interval		
			Point Estimate	Lower	Upper	
Pair 1	What is your overall perception of brand (Uber)? - Brand Perception	Cohen's d	1.748	-2.130	-2.536	-1.720
	On a scale from 0-10, how would you rate your most recent Uber ride?	Hedges' correction	1.766	-2.109	-2.510	-1.703

a. The denominator used in estimating the effect sizes.

Cohen's d uses the sample standard deviation of the mean difference.

Hedges' correction uses the sample standard deviation of the mean difference, plus a correction factor.



SPSS Model 2 (Slide 11): Paired Samples t-Test

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	What is your overall perception of brand (Uber)? - Brand Perception	3.63	86	1.052	.113
	On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague?	6.80	86	2.548	.275

Paired Samples Correlations

		N	Correlation	Significance	
				One-Sided p	Two-Sided p
Pair 1	What is your overall perception of brand (Uber)? - Brand Perception & On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague?	86	.793	<.001	<.001

Paired Samples Test

		Paired Differences				Significance			
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference		t	df	
					Lower	Upper			One-Sided p
Pair 1	What is your overall perception of brand (Uber)? - Brand Perception - On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague?	-3.174	1.829	.197	-3.567	-2.782	-16.094	85	<.001
									Two-Sided p
									<.001

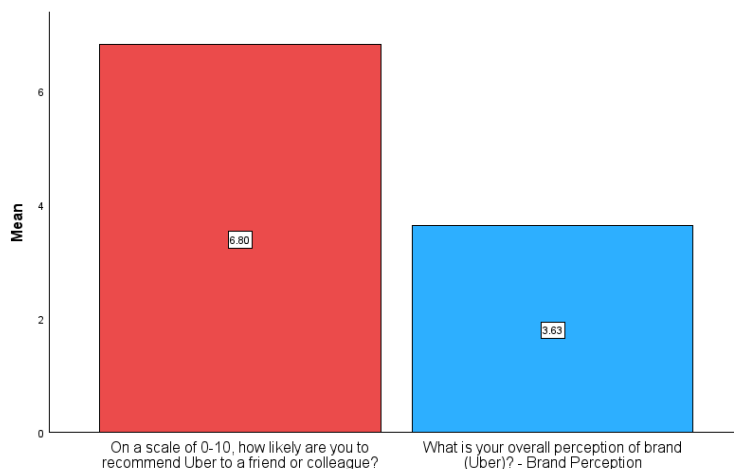
Paired Samples Effect Sizes

		Standardizer ^a		Point Estimate	95% Confidence Interval	
					Lower	Upper
Pair 1	What is your overall perception of brand (Uber)? - Brand Perception	Cohen's d	1.829	-1.735	-2.069	-1.398
	On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague?	Hedges' correction	1.845	-1.720	-2.050	-1.386

a. The denominator used in estimating the effect sizes.

Cohen's d uses the sample standard deviation of the mean difference.

Hedges' correction uses the sample standard deviation of the mean difference, plus a correction factor.



SPSS Model 3 (Slide 12): Regression Analysis

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.803 ^a	.645	.589	.625

a. Predictors: (Constant), Which device do you usually use to request an Uber? – Selected Choice, How do you describe your gender?, What is the highest level of education you have completed?, What was your total household income (in USD) before taxes in the past 12 months?, How willing are you to pay more for Uber compared to other transportation options if it provides greater convenience or quality? – Fare Cost, On a scale of 0–10, how likely are you to recommend Uber to a friend or colleague? – Group, What is your age?, On average, how many Uber rides do you take in a typical week?, On a scale from 0–10, how would you rate your most recent Uber ride? – Group

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	40.422	9	4.491	11.514	<.001 ^b
	Residual	22.235	57	.390		
	Total	62.657	66			

a. Dependent Variable: What is your overall perception of brand (Uber)? – Brand Perception

b. Predictors: (Constant), Which device do you usually use to request an Uber? – Selected Choice, How do you describe your gender?, What is the highest level of education you have completed?, What was your total household income (in USD) before taxes in the past 12 months?, How willing are you to pay more for Uber compared to other transportation options if it provides greater convenience or quality? – Fare Cost, On a scale of 0–10, how likely are you to recommend Uber to a friend or colleague? – Group, What is your age?, On average, how many Uber rides do you take in a typical week?, On a scale from 0–10, how would you rate your most recent Uber ride? – Group

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.258	.676		1.862	.068
	On a scale of 0–10, how likely are you to recommend Uber to a friend or colleague? – Group	.797	.157	.554	5.091	<.001
	On average, how many Uber rides do you take in a typical week?	.292	.222	.119	1.318	.193
	On a scale from 0–10, how would you rate your most recent Uber ride? – Group	.133	.145	.101	.917	.363
	How willing are you to pay more for Uber compared to other transportation options if it provides greater convenience or quality? – Fare Cost	.210	.078	.226	2.704	.009
	What is your age?	.196	.088	.189	2.218	.031
	How do you describe your gender?	-.172	.132	-.110	-1.302	.198
	What is the highest level of education you have completed?	-.007	.092	-.006	-.073	.942
	What was your total household income (in USD) before taxes in the past 12 months?	-.019	.024	-.067	-.786	.435
	Which device do you usually use to request an Uber? – Selected Choice	-.509	.187	-.232	-2.716	.009

a. Dependent Variable: What is your overall perception of brand (Uber)? – Brand Perception

Slide 14 (Brand Perception and Likelihood to Recommend)

Oneway

		ANOVA				
		Sum of Squares	df	Mean Square	F	Sig.
How long have you been an Uber member?	Between Groups	11.244	4	2.811	1.721	.154
	Within Groups	130.708	80	1.634		
	Total	141.953	84			
How often have you used Uber in the past 6 months?	Between Groups	2.420	4	.605	.508	.730
	Within Groups	96.510	81	1.191		
	Total	98.930	85			
On average, how many Uber rides do you take in a typical week?	Between Groups	.151	4	.038	.209	.933
	Within Groups	14.651	81	.181		
	Total	14.802	85			
On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague? - Group	Between Groups	20.909	4	5.227	20.482	<.001
	Within Groups	20.672	81	.255		
	Total	41.581	85			
On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague?	Between Groups	375.764	4	93.941	43.265	<.001
	Within Groups	175.875	81	2.171		
	Total	551.640	85			

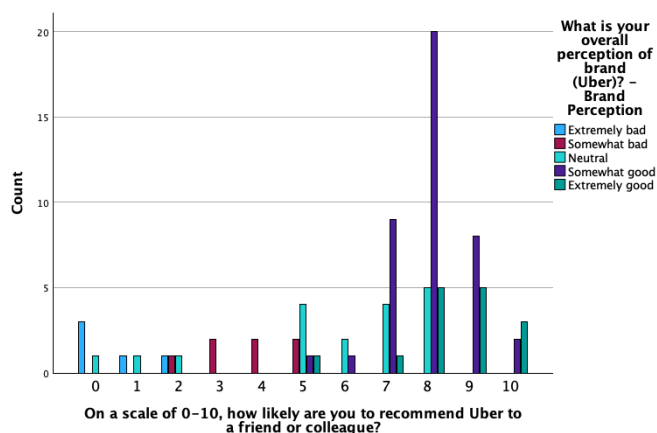
ANOVA Effect Sizes^{a,b}

		Point Estimate	95% Confidence Interval	
			Lower	Upper
How long have you been an Uber member?	Eta-squared	.079	.000	.169
	Epsilon-squared	.033	-.050	.128
	Omega-squared Fixed-effect	.033	-.049	.127
	Omega-squared Random-effect	.008	-.012	.035
How often have you used Uber in the past 6 months?	Eta-squared	.024	.000	.072
	Epsilon-squared	-.024	-.049	.026
	Omega-squared Fixed-effect	-.023	-.049	.026
	Omega-squared Random-effect	-.006	-.012	.007
On average, how many Uber rides do you take in a typical week?	Eta-squared	.010	.000	.026
	Epsilon-squared	-.039	-.049	-.022
	Omega-squared Fixed-effect	-.038	-.049	-.022
	Omega-squared Random-effect	-.009	-.012	-.005
On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague? - Group	Eta-squared	.503	.324	.597
	Epsilon-squared	.478	.291	.577
	Omega-squared Fixed-effect	.475	.288	.575
	Omega-squared Random-effect	.185	.092	.252
On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague?	Eta-squared	.681	.547	.745
	Epsilon-squared	.665	.524	.732
	Omega-squared Fixed-effect	.663	.521	.730
	Omega-squared Random-effect	.330	.214	.403

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

b. Negative but less biased estimates are retained, not rounded to zero.

Graph



Slide 15 - 16 (Types of Device Used and App Issues)

Warnings

Post hoc tests are not performed for How frequently do you experience the following app issues? – Difficulty Navigating within App because at least one group has fewer than two cases.

Post hoc tests are not performed for How frequently do you experience the following app issues? – Error Messages because at least one group has fewer than two cases.

ANOVA

		Sum of Squares	df	Mean Square	F	Sig.
How frequently do you experience the following app issues? – Difficulty Navigating within App	Between Groups	7.538	2	3.769	5.354	.007
	Within Groups	53.500	76	.704		
	Total	61.038	78			
How frequently do you experience the following app issues? – Error Messages	Between Groups	8.094	2	4.047	6.454	.003
	Within Groups	47.653	76	.627		
	Total	55.747	78			

ANOVA Effect Sizes^{a,b}

		Point Estimate	95% Confidence Interval	
			Lower	Upper
How frequently do you experience the following app issues? – Difficulty Navigating within App	Eta-squared	.123	.011	.253
	Epsilon-squared	.100	-.015	.233
	Omega-squared Fixed-effect	.099	-.015	.231
	Omega-squared Random-effect	.052	-.007	.131
How frequently do you experience the following app issues? – Error Messages	Eta-squared	.145	.021	.278
	Epsilon-squared	.123	-.005	.259
	Omega-squared Fixed-effect	.121	-.005	.256
	Omega-squared Random-effect	.065	-.002	.147

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

b. Negative but less biased estimates are retained, not rounded to zero.

Frequencies

Statistics

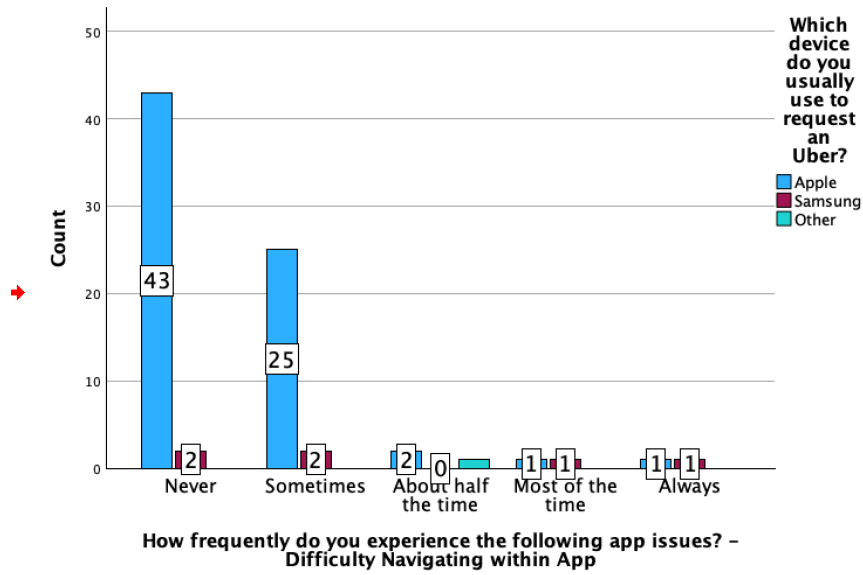
Which device do you usually use to request an Uber?

N	Valid	79
	Missing	11

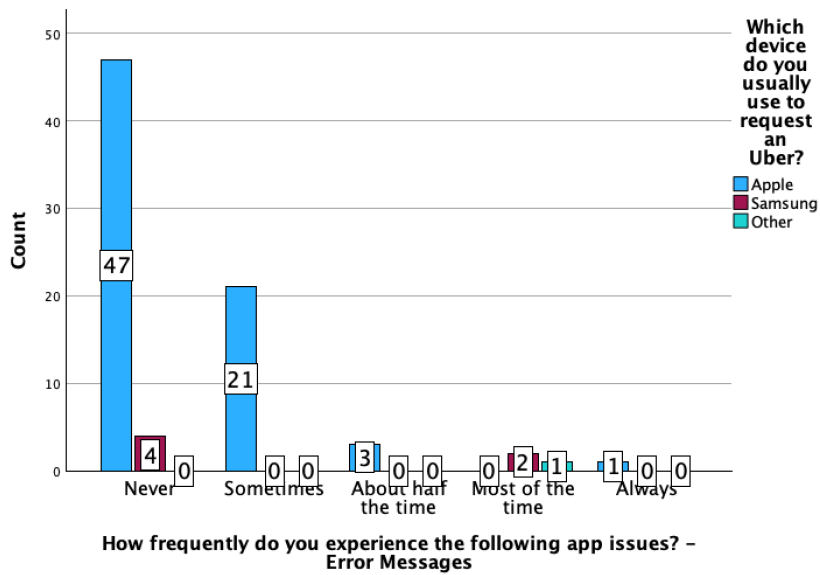
Which device do you usually use to request an Uber?

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Apple	72	80.0	91.1	91.1
	Samsung	6	6.7	7.6	98.7
	Other	1	1.1	1.3	100.0
	Total	79	87.8	100.0	
Missing	System	11	12.2		
Total		90	100.0		

Graph



Graph



Slide 17 (Income and Type of Uber Chosen)

Oneway

ANOVA

What type of Uber ride do you usually choose?

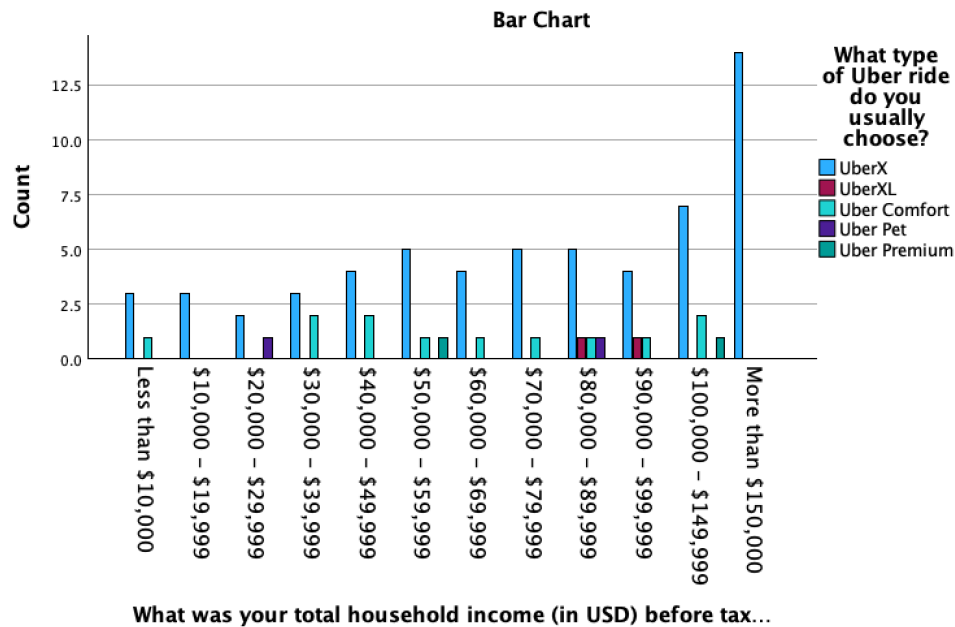
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	11.249	11	1.023	.695	.738
Within Groups	95.608	65	1.471		
Total	106.857	76			

ANOVA Effect Sizes^{a,b}

		Point Estimate	95% Confidence Interval	
			Lower	Upper
What type of Uber ride do you usually choose?	Eta-squared	.105	.000	.110
	Epsilon-squared	-.046	-.169	-.041
	Omega-squared Fixed-effect	-.046	-.167	-.040
	Omega-squared Random-effect	-.004	-.013	-.004

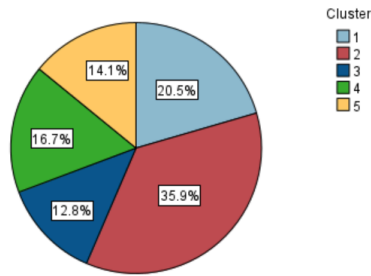
a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

b. Negative but less biased estimates are retained, not rounded to zero.



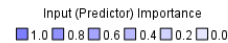
Cluster Analysis (Slide 18 - 21)

Cluster Sizes



Size of Smallest Cluster	10 (12.8%)
Size of Largest Cluster	28 (35.9%)
Ratio of Sizes: Largest Cluster to Smallest Cluster	2.80

Clusters



Cluster	1	2	3	4	5
Label					
Description					
Size	20.5% (16)	35.9% (28)	12.8% (10)	16.7% (13)	14.1% (11)
Inputs	On average, how many Uber rides do you take in a typical week? 2.00 What is your overall perception of brand (Uber)? - Brand Perception Somewhat good On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague? 7.81 How often have you used Uber in the past 6 months? 3.44 What is your age? 2.88 What is the highest level of education you have completed? 3.38 How long have you been an Uber member? 3.12 Which device do you usually use to request an Uber? 1.00 How do you describe your gender? 1.31	On average, how many Uber rides do you take in a typical week? 1.00 What is your overall perception of brand (Uber)? - Brand Perception Somewhat good On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague? 8.00 How often have you used Uber in the past 6 months? 2.04 What is your age? 2.93 What is the highest level of education you have completed? 2.64 How long have you been an Uber member? 4.00 Which device do you usually use to request an Uber? 1.00 How do you describe your gender? 1.64	On average, how many Uber rides do you take in a typical week? 1.00 What is your overall perception of brand (Uber)? - Brand Perception Extremely good On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague? 8.30 How often have you used Uber in the past 6 months? 2.70 What is your age? 4.20 What is the highest level of education you have completed? 2.20 How long have you been an Uber member? 4.30 Which device do you usually use to request an Uber? 1.10 How do you describe your gender? 1.70	On average, how many Uber rides do you take in a typical week? 1.00 What is your overall perception of brand (Uber)? - Brand Perception Neutral (100.0%) On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague? 6.00 How often have you used Uber in the past 6 months? 1.85 What is your age? 3.54 What is the highest level of education you have completed? 2.31 How long have you been an Uber member? 3.00 Which device do you usually use to request an Uber? 1.31 How do you describe your gender? 1.54	On average, how many Uber rides do you take in a typical week? 1.18 What is your overall perception of brand (Uber)? - Brand Perception Somewhat bad On a scale of 0-10, how likely are you to recommend Uber to a friend or colleague? 2.64 How often have you used Uber in the past 6 months? 2.27 What is your age? 3.18 What is the highest level of education you have completed? 3.18 How long have you been an Uber member? 4.00 Which device do you usually use to request an Uber? 1.36 How do you describe your gender? 2.09